

Question	Marking Guidance	Mark	Comments
09.1	1. A <u>metabolite</u> in condensation/hydrolysis/ photosynthesis/respiration; 2. A solvent so (metabolic) reactions can occur OR A solvent so allowing transport of substances; 3. High (specific) heat capacity so buffers changes in temperature; 4. Large latent heat of vaporisation so provides a cooling effect (through evaporation); 5. Cohesion (between water molecules) so supports columns of water (in plants); 6. Cohesion (between water molecules) so produces surface tension supporting (small) organisms;	5 max	3. For 'buffer' accept 'resist'. 4. Reject latent heat of evaporation 5. For 'columns of water' accept 'transpiration stream'. Do not credit 'transpiration' alone but accept description of 'stream'. 5. For 'columns of water' accept 'cohesion-tension (theory)'. 5. and 6. For cohesion accept hydrogen bonding Ignore reference to pH. Allow other suitable properties but must have a valid explanation. For example • ice floating so maintaining aquatic habitat beneath • water transparent so allowing light penetration for photosynthesis

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09.2	1. DNA helicase unwinds DNA/double helix OR DNA helicase breaks hydrogen bonds; 2. Both strands act as templates; 3. (Free DNA) nucleotides line up in complementary pairs/A-T and G-C; 4. DNA polymerase joins nucleotides (of new strand); 5. Forming phosphodiester bonds; 6. Each new DNA molecule consists of one old/original/template strand and one new strand;	5 max	2. Accept description of 'template', eg exposed bases on single (polynucleotide) strands 4. Reject forms hydrogen bonds/joins bases
TOTAL		10	